

COURSE HANDOUT (PART-II)

In addition to Part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : BIO F311
Course Title : RECOMBINANT DNA TECHNOLOGY
Instructor-in-charge : Prof. SOUMITRA GHOSH
Instructor : Prof. Vishal Saxena
Tutorial Instructors : Prof. Soumitra Ghosh & Prof. Vishal Saxena

1. Course Description:

The course deals with theoretical aspects and selected experiments of recombinant DNA manipulation. Emphasis will be placed on procedures to create chimeric molecules using examples from actual experimental work. Vector designing, PCR, DNA sequencing, *in vitro* mutagenesis, cloning in prokaryotic and eukaryotic systems and whole genome approaches will be covered.

2. Scope and Objective of the course:

Almost every field of modern experimental biology relies heavily on DNA manipulation techniques. This course will expose the student to the foundations and principles underlying such molecular biology methods and give a brief yet broad overview of the recent trends and applications of recombinant DNA technology.

3. Text Book (TB):

Howe, Christopher J. Gene Cloning and Manipulation (2nd edition). New Delhi: Cambridge University Press India Pvt. Ltd., 2015.

4. Reference Books (RB):

- **RB_a**: Brown, Terence A. Gene Cloning and DNA Analysis: An Introduction (8th edition). United Kingdom: Wiley-Blackwell, 2010.
- **RB_b**: Primrose, Sandy B. and Richard Twyman. Principles of Gene Manipulation and Genomics (7th edition). New York: Blackwell Science, 2006.

5. Course Plan:

MODULE	LECTURE TOPICS	TB/ RB CHAP#	LEARNING OUTCOMES
I Orientation to the course. Tools and techniques of recombinant DNA technology: DNA manipulation Enzymes and techniques	L1: Defining the scope and context of the course. Introduction to Recombinant DNA Techniques and workflow summary.	TB 1-2 TB 5-6	Understanding of basic tools/ techniques of Recombinant DNA Technology; Restriction endonucleases; Understanding of Polymerase chain reaction with variants.
	L2-5: Enzymes for manipulating DNA; Restriction mapping and analysis		
	L6-12: Understanding PCR; different types of PCR – inverse PCR, multiplex PCR, quantitative real time PCR, DD PCR Genomic, cDNA & specialized libraries		



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II Techniques for Cloning and Recombinant DNA Analysis Cloning vectors, Electrophoresis Protein expression	L13-19: Vectors for cloning - Plasmids, DNA isolation procedure; bacterial host and its transformation; High-capacity cloning vectors (BAC, PAC, etc.); bacteriophages (including molecular aspects of phages pertinent to vector design); Viral vectors, Expression vectors, fusion tags	TB 1 TB 3-6 Class Notes	Understanding of how proteins are expressed in <i>E. coli</i> . Expression and detection of recombinant proteins.
	L20-22: Electrophoresis of nucleic acids; Various methods used for screening and selection of the clones;		
	L23-25: Protein expression, SDS PAGE, Western Blotting, Immunofluorescence imaging		
	L26-29: Nucleic acid hybridization techniques- Southern, Northern blotting, FISH		
III Techniques for Mutagenesis & Genome editing	L30-34: Methods for site-directed mutagenesis, Genome editing by CRISPR-Cas and related gene modification techniques.	TB 7 Class Notes	Knowledge to base specific gene sequence modification and genome editing
IV DNA sequencing	L35-40: Sanger's chemistry; NGS automated sequencing; latest advances	TB 1 RB _a 10 Class Notes	Understanding of gene sequencing and the concepts behind making of NGS libraries.
V Applications of recombinant DNA technology	L41-43: Selected applications of recombinant methodologies in medicine, agriculture & biomedical science, e.g., gene therapy, RNAi, etc.	TB 8 TB 9	Some latest advances in this area (assignments)

6. Evaluation Scheme:

No.	Evaluation component	Duration	Date and time	(%) Weightage	Remarks
1.	Mid-Semester Exam	1.5 hrs	05/10/2026 (FN1)	35%	Closed Book
2.	Quizzes/ Assignments	Variable	Pre-announced, and surprise activities in class/tutorial hours.	25%	Will vary as per component
3.	Comprehensive Exam	3 hrs	01/12/ 2026 (FN)	40%	Closed/ Open Book

7. **Grading Policy:** Award of grades would be guided in general by the histogram of marks. Decision for borderline cases would also take into consideration the student's attendance and participation in classes. If a student scores below 30% of the average of the top three performers or 40% of median (whichever is lower), a 'NC' (Not Cleared) grade may be awarded.

8. **Chamber Consultation:** By appointment through email.

9. **Make-up Policy:** Makeup will be granted only for Mid-Semester or Comprehensive examinations as per AUGSD guidelines with compulsorily a request email copy to the instructors. There shall be no makeups for any announced or unannounced quizzes or assignments.

10. **Notices:** All notifications will be sent through email.

Instructor-in-Charge (BIOF311)



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